Limit Order Book Research

从微观结构到可执行交易信号

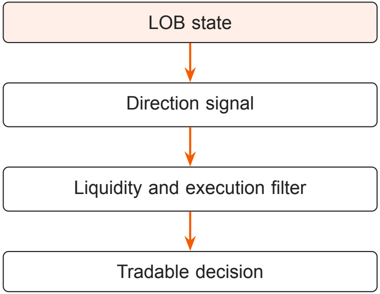
15-minute research briefing | June 4, 2026

Executive Takeaway

一句话结论

LOB 研究的价值不只是预测下一跳价格，而是把市场状态拆成 **方向、流动性、可成交性、风险约束**，判断信号能否转化为真实 P&L。

- 短周期价格预测容易被 spread、impact、queue position 吃掉。
- 订单簿状态提供比价格序列更接近交易执行的状态变量。
- 内网数据应重点验证：信号方向是否稳定、流动性是否足够、执行成本是否可控。

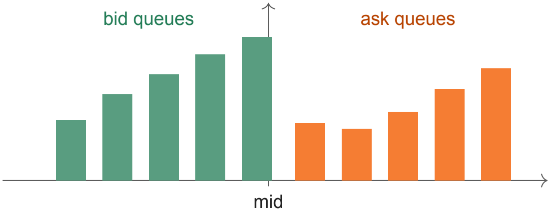


What a LOB State Contains

- 每个价位有 price 和 queue size：$bid_i, ask_i, q_i^b, q_i^a$。
- 常用输入是最近 T 个时刻、前 L 档订单簿：

$$X_t \in \mathbb{R}^{T\times 4L}.$$

- 关键不是一根价格线，而是 bid/ask 两侧队列随时间的形状变化。



近端不平衡

$$I_t^{(k)} = \frac{\sum_{i=1}^{k} q_i^b - \sum_{i=1}^{k} q_i^a}{\sum_{i=1}^{k} q_i^b + \sum_{i=1}^{k} q_i^a}$$

Price move intuition

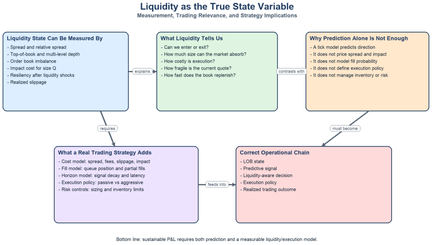
当 best bid/ask 队列被 market orders 和 cancellations 消耗完，价格跳到下一档。

Why Liquidity Is the True State Variable

- Direction tells us where price may move.
- Liquidity tells us whether the move can be monetized.
- Execution converts a prediction into realized P&L.

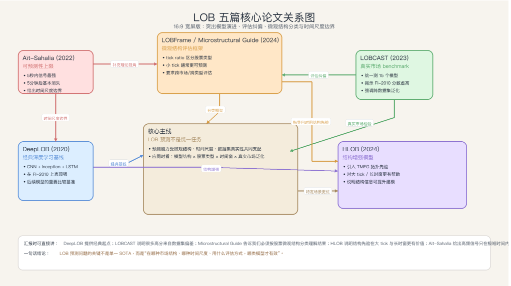
Decision state

$$s_t = (price,\ book,\ liquidity,\ execution,\ risk).$$



Research Landscape

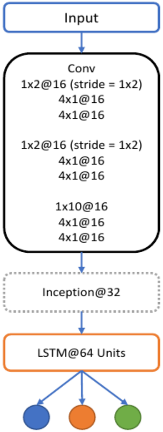
- Queueing models: explain price moves through queue depletion.
- Hawkes processes: capture self-exciting and cross-exciting order flow.
- DeepLOB: learns local spatial structure and temporal dependency.
- LOBFrame / newer benchmarks: emphasize fair evaluation and market microstructure constraints.



Source: Cont and de Larrard; Bacry et al.; Zhang et al.; recent LOB benchmark and guide papers.

Modeling Pattern: DeepLOB as a Useful Baseline

- CNN learns local price-level and bid/ask interaction patterns.
- Inception-style layers capture different local scales.
- LSTM/sequence block captures recent temporal dynamics.
- Output is usually a short-horizon up/down/stationary label.



Practical message

Deep model matters less than having correct labels, leak-free evaluation, and executable cost-aware metrics.

From Prediction to Trading Signal

A signal is tradable only if

$$E[\Delta p] > spread + fee + impact + delay\ risk.$$

- Direction: expected sign and confidence.
- Liquidity: depth, spread, imbalance, resiliency.
- Execution: aggressive/passive choice, queue priority, slippage.
- Risk: holding horizon, inventory, adverse selection.

Raw model score → Liquidity filter → Execution-cost filter → Position sizing → Trade / no trade

Internal Data: What We Need to Validate

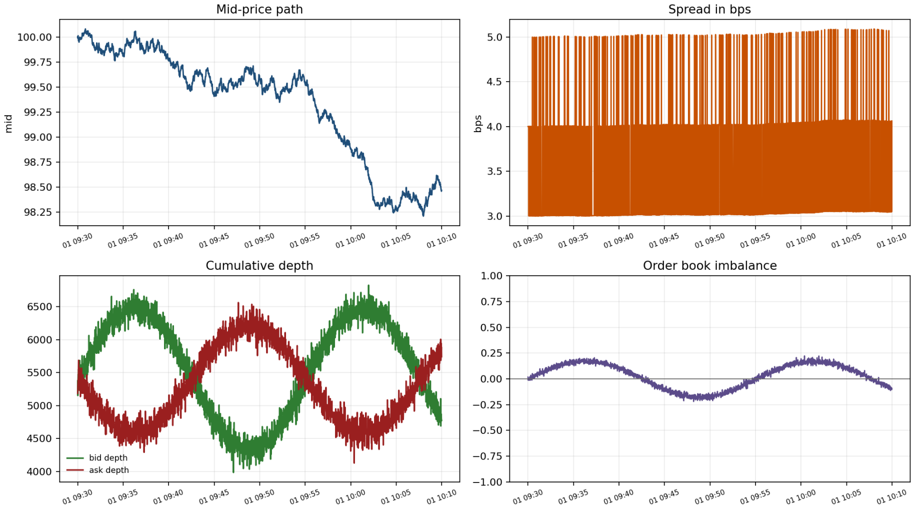
- Schema: timestamp, first L bid/ask prices and sizes, trades, cancellations if available.
- Metrics: spread, depth, imbalance, impact proxy, resiliency.
- Label: future mid-price movement at 10s/30s/1min or event horizon.
- Evaluation: time split, symbol split, no leakage, cost-aware P&L simulation.

Current data source

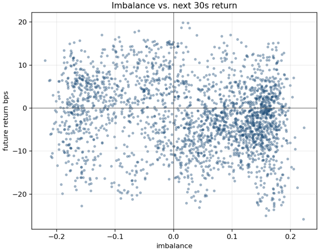
Rows	2,400
Median spread	3.05 bps
Median depth	10,865
Impact proxy	1.62 bps
Imb-ret corr	-0.166

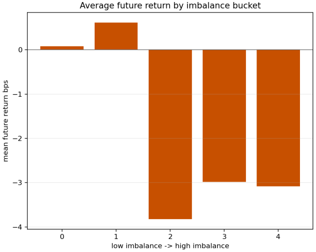
Data source: synthetic placeholder; replace with intranet data

Internal Data Snapshot



Signal Diagnostic





How to read this

- If imbalance has no stable relationship with future return, we should not trade it directly.
- If it works only under tight spread/deep book regimes, it should be a conditional signal.
- The final test is execution-aware P&L, not classification accuracy.

Recommended Research Workflow

1. Build clean LOB snapshots and event-aligned labels.
2. Start with interpretable features: spread, depth, imbalance, impact proxy.
3. Benchmark simple models before DeepLOB.
4. Add execution filter and cost-aware backtest.
5. Only then test deep sequence models.

Minimum viable internal study

- 3–5 liquid instruments.
- Multiple market regimes: open, normal, volatile, close.
- Report: signal IC, turnover, cost, realized slippage, capacity.

Roadmap and Ask

Next two weeks

- Week 1: internal data quality, feature library, baseline diagnostics.
- Week 2: cost-aware evaluation, signal filters, first strategy prototype.
- Optional: DeepLOB/Transformer benchmark after baseline is stable.

Decision needed

- Which instruments and venues should be prioritized?
- What execution assumption is acceptable for the first backtest?
- What capacity threshold makes the signal commercially relevant?

LOB research should be evaluated as an execution problem, not only a prediction problem.